# PCA Component Sufficiency Analysis for High-Fidelity Approximation of OFIQ Unified Face Image Quality Scores

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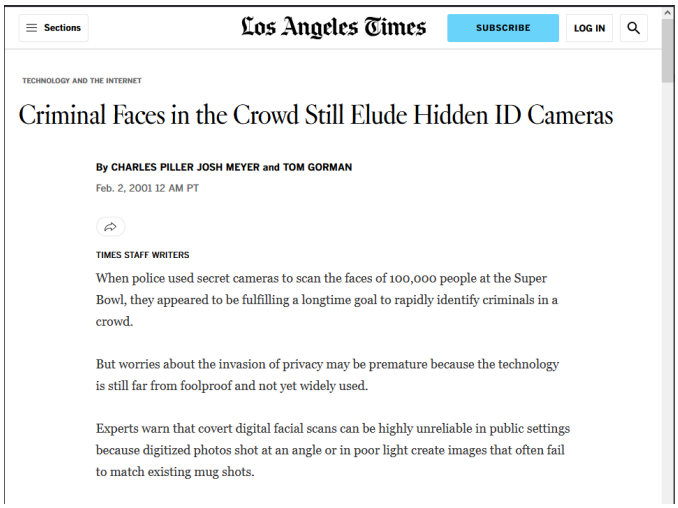
***Abstract*—The Open-Source Face Image Quality (OFIQ) framework is the most concise and operationally relevant summary of face image appropriateness for biometric recognition tasks. It offers a full array of scalar quality indicators in addition to a single intrinsic unified quality score. However, large-scale and resource-constrained deployments face practical difficulties due to the computational expense of performing the entire OFIQ process. This has prompted research into lightweight approximation techniques that can accurately replicate the unified quality score. To determine the minimal number of principal components needed to achieve reconstruction accuracy at or above 95% agreement with the original score, this study examines the use of Principal Component Analysis as a primary data-driven approximation technique for the OFIQ unified quality score. As a secondary comparative analysis, Independent Component Analysis is applied in parallel to the same training and validation data to assess whether a statistically independent decomposition of the OFIQ metric space produces a higher-fidelity approximation than the variance-based PCA approach.**

**Instead of imposing a fixed component count, a variance threshold approach is used for PCA, in which components are gradually kept until the PCA model's cumulative explained variance surpasses a predetermined sufficiency threshold, while ICA employs a fixed component count determined by the structure of the metric space. Both models are trained on face image quality metrics extracted from three large-scale and demographically heterogeneous datasets — VGGFace2, Flickr Faces HQ, and CelebA — and applied to held-out validation partitions to produce composite quality scores. Reconstruction fidelity is assessed by computing Pearson correlation, R-squared, and mean absolute error between each method's composite score and the OFIQ unified quality score, with results compared across methods to determine which decomposition approach more faithfully approximates the reference output.**

## I. Introduction

Due to its passive collection, scalability, and interoperability with current consumer and surveillance infrastructure, facial recognition has become the most widely implemented of the many modalities used in contemporary biometric systems. Its incorporation into commercial, governmental, security, and personal applications has grown significantly during the last 20 years. For the purposes of this study, face recognition refers to a computational system's ability to recognize or confirm a person's identification by analyzing distinguishing facial traits taken from pictures or videos.



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TECHNOLOGY AND THE INTERNET

Criminal Faces in the Crowd Still Elude Hidden ID Cameras

By CHARLES PILLER JOSH MEYER and TOM GORMAN

Feb. 2, 2001 12 AM PT

TIMES STAFF WRITERS

When police used secret cameras to scan the faces of 100,000 people at the Super Bowl, they appeared to be fulfilling a longtime goal to rapidly identify criminals in a crowd.

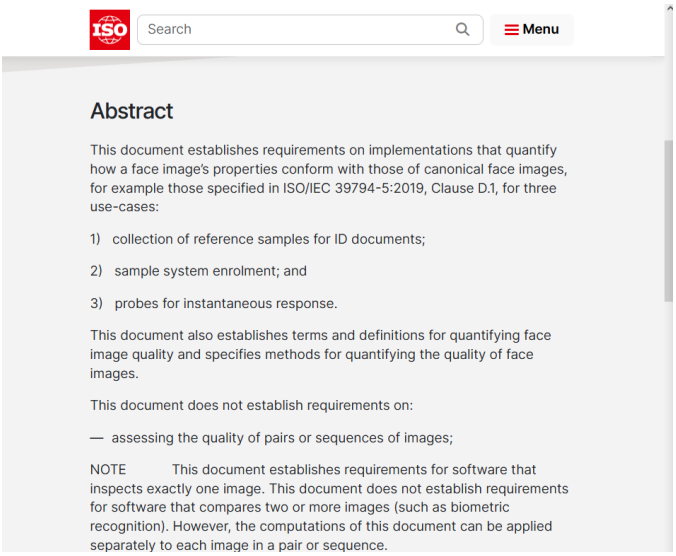
But worries about the invasion of privacy may be premature because the technology is still far from foolproof and not yet widely used.

Experts warn that covert digital facial scans can be highly unreliable in public settings because digitized photos shot at an angle or in poor light create images that often fail to match existing mug shots.

Fig. 1. Florida Super Bowl Viisage Technologies LA Times Article

At the 2001 Super Bowl in Tampa, Florida, facial recognition software created by Viisage Technology was used to screen over 100,000 attendees against a criminal mugshot database in real time — one of the most striking early examples of this technology's potential. This in part helped to bring facial recognition into the spotlight, as noted by the large amount of articles at the time reporting on this phenomena, like the LA Times article in figure 1. This feat would have been logistically impossible using traditional manual identification methods. Even though limitations like a 43% false rejection rate highlighted the substantial room for advancement that has driven the field ever since, the system's ability to successfully identify 19 people with outstanding warrants in a crowd of that size highlighted the transformational promise of automated facial biometrics as a practical identification tool [1].

Facial recognition systems function within larger multimodal biometric architectures in modern deployments, which calls for the creation of interoperability standards to guar-



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## Abstract

This document establishes requirements on implementations that quantify how a face image's properties conform with those of canonical face images, for example those specified in ISO/IEC 39794-5:2019, Clause D.1, for three use-cases:

1) collection of reference samples for ID documents;

2) sample system enrolment; and

3) probes for instantaneous response.

This document also establishes terms and definitions for quantifying face image quality and specifies methods for quantifying the quality of face images.

This document does not establish requirements on:

— assessing the quality of pairs or sequences of images;

NOTE This document establishes requirements for software that inspects exactly one image. This document does not establish requirements for software that compares two or more images (such as biometric recognition). However, the computations of this document can be applied separately to each image in a pair or sequence.

Fig. 2. Official Website for ISO/IEC 29794-5

antee dependable and consistent performance across diverse platforms. The most important of these is ISO/IEC 29794-5, which specifically aims to guarantee consistency and alignment across biometric systems industry-wide by establishing standardized quality measures for facial images [2]. The main website for this standard can be seen in figure 2. According to this paradigm, characteristics like brightness, occlusion, and sharpness are acknowledged as key factors that determine the quality of facial images and have a direct impact on the precision and dependability of subsequent recognition processes. This has significant practical implications because it has been demonstrated that low-quality imagery in training or inference pipelines directly reduces recognition accuracy and introduces systemic bias at scale. In response, frameworks for open-source quality assessment were created, such as the Open-Source Face Image Quality (OFIQ) system, to offer a consistent, easily accessible, and compatible method for evaluating the quality of facial images. OFIQ, first published in 2020, is the official reference implementation of ISO/IEC 29794-5 and supports facial image curation and quality-gated acquisition across biometric systems in both US and EU contexts by offering standardized scalar quality metrics and a single intrinsic unified quality score [3].

OFIQ plays a fundamental role in guaranteeing that facial recognition systems supporting contemporary security, border control, and consumer identification use data of adequate and consistent quality by providing a standard quality language across platforms and institutions. The computing requirements of operating the entire OFIQ pipeline pose significant practical challenges despite its substantial benefit, especially when applied to the large-scale datasets typical of real-world biometric deployments. The framework's extensive metric output, which includes dozens of scalar quality indicators calculated for each image, adds significant processing cost that limits its applicability in situations with limited resources and real-time constraints. OFIQ's intrinsic unified quality score — a single scalar value that condenses the whole scope of the framework's quality assessment into one operationally useful output — is particularly interesting in this context. Large-scale biometric pipelines would experience significant efficiency gains if this unified score could be accurately approximated using a much smaller collection of computationally derived features.

Principal Component Analysis (PCA) is an established and mathematically rigorous candidate for this purpose, offering the capacity to identify the axes of maximum variance within a high-dimensional feature space and represent that space in a compact, lower-dimensional form without significantly losing informational content. Independent Component Analysis (ICA) presents a complementary alternative, seeking to recover statistically independent latent factors underlying the observed metrics rather than axes of maximum variance — an assumption that may better reflect the structure of facial image quality if its underlying drivers are genuinely independent.

This study pursues two parallel objectives. The primary objective is to establish the minimum number of principal components required to achieve reconstruction fidelity at or above 95% cumulative explained variance when approximating the OFIQ unified quality score using PCA. The secondary objective is to apply FastICA to the same training and validation data in parallel, using a fixed component count, and evaluate whether the resulting ICA composite score more faithfully approximates the OFIQ unified quality score than the PCA composite. Both methods are evaluated using Pearson correlation, R-squared, and mean absolute error against the OFIQ unified quality score across the combined validation set. Section II reviews related work on dimensionality reduction and dataset characteristics. Section III details the methodology encompassing both PCA and ICA formulations. Section IV presents the full experimental setup. Section V reports and analyzes the comparative results. Section VI contains all references.

## II. Related Works

### A. Data Dimensional Reduction and Principal Components Analysis [4]

Salem and Hussein examine the theoretical underpinnings and practical applications of Principal Component Analysis for reducing the dimensionality of high-dimensional datasets while maintaining the most analytically significant structure of the original variable space.

The study's motivation stems from the observation that the analysis of inter-variable relationships and the visualization of structural patterns become increasingly difficult using traditional analytical techniques as datasets increase in size and dimensionality. Such methods are impractical for large datasets and require a more rigorous analytical framework, since the number of bivariate plots needed to thoroughly analyze pairwise correlations in a dataset of n variables scales conjunctively. To address this, the paper introduces PCA as a mathematically rigorous unsupervised learning method that can reduce a set of potentially correlated variables to a smaller

set of uncorrelated principal components, each arranged according to the percentage of total variance it extracts from the original feature space. The authors provide a comprehensive explanation of the mathematical foundations of PCA by standardizing variables, creating the covariance matrix, and using eigen decomposition to determine the principal component directions. Importantly, the study also demonstrates how PCA and Singular Value Decomposition are related, showing that both methods produce equivalent principal component solutions and that SVD provides computational benefits for large-scale implementations.

Using a real-world implementation on the Iris dataset, the authors validate their theoretical treatment, reducing a four-dimensional feature space to a two-dimensional principal component subspace. Their findings show that the first two principal components alone account for 99.96% of the dataset's total variance, demonstrating that when components are chosen based on cumulative variance contribution, a significantly smaller component set can preserve almost all of the original feature space's informational content.

This work is directly and significantly relevant to the current investigation, providing theoretical and methodological guidance for both the PCA and ICA analyses. The variance threshold approach used in this study to determine the minimum number of PCA components sufficient to approximate the OFIQ unified quality score is directly supported by the formal treatment of how principal components are constructed as linear combinations of original variables that maximize captured variance. The preprocessing approach used here — specifically centering variables to zero mean and unit variance prior to decomposition — is equally essential for ICA, as FastICA requires standardized inputs to converge reliably. Furthermore, Salem and Hussein's demonstration that a small number of components can account for an overwhelming proportion of total dataset variance provides empirical support for the central premise shared by both methods in this study: that a compact decomposition of OFIQ's scalar metric space can serve as a high-fidelity approximation of its unified quality score.

### *B. Effects of Dataset Features on the Performance of Classification Algorithms [5]*

Kwon and Sim conduct a methodical empirical study examining how intrinsic dataset properties affect classification algorithms' accuracy and computational efficiency under various circumstances. Rather than proposing a novel algorithm, the authors focus on systematically identifying the quantifiable dataset attributes that have the greatest influence on classification performance. Using 106 benchmark datasets from the UCI repository spanning significant variation in sample size, feature dimensionality, class distribution balance, attribute type, missing value prevalence, and functional dependency among features, the study evaluates eight representative supervised classification algorithms using a multiple regression framework.

The results show that data imbalance is the most consistently significant factor, statistically lowering classification accuracy in all evaluated methods, while high feature dimensionality in relation to sample size also emerged as a strong negative predictor of performance across most classifiers. Datasets with balanced class distributions, reduced feature redundancy, and sufficient sample depth consistently yielded more accurate and stable model outputs regardless of algorithm selection. The study concludes that dataset structure has an impact on algorithmic performance equal to, and sometimes greater than, the algorithm selection itself.

This work carries direct implications for both the PCA and ICA analyses in the present study. The finding that feature dimensionality and data imbalance are the primary determinants of model performance degradation motivates the deliberate selection of three structurally diverse datasets — VGGFace2, FFHQ, and CelebA — that collectively expose both decomposition methods to the range of distributional variation that Kwon and Sim identify as most consequential for model stability. Crucially, this framework also provides a basis for interpreting any performance difference between PCA and ICA across the validation set — divergence in approximation fidelity between the two methods may be attributable not to an inherent superiority of one decomposition approach but to differing sensitivity to the structural properties of the combined dataset, including its dimensionality relative to sample size and the degree of feature redundancy present in the OFIQ scalar metric space.

## III. Methodology

The methodology of this study is structured around four primary objectives: implementing a variance-threshold-based PCA formulation capable of approximating the OFIQ unified quality score with high fidelity; implementing a parallel FastICA formulation on the same data to serve as a comparative secondary analysis; training and validating both formulations across a pooled combination of three large-scale and demographically diverse facial image datasets; and evaluating the composite quality scores produced by each method against the OFIQ unified quality score using multiple complementary performance metrics. The subsections that follow describe each component of the pipeline in full.

### *A. Datasets*

Three large-scale facial image datasets are employed in this study, selected for their collectively broad coverage of identity, demographic composition, acquisition environment, image resolution, and capture condition diversity. VGGFace2 comprises over 3.3 million images spanning more than 9,000 unique identities with significant variation in pose, age, ethnicity, and illumination. Flickr Faces HQ (FFHQ) contains 70,000 high-resolution images sourced from Flickr, providing considerable diversity in age, ethnicity, and image quality. CelebA consists of over 200,000 celebrity facial images annotated with 40 binary attributes, encompassing a wide range of facial appearances and acquisition settings.

Images from all three datasets are pooled into a single combined corpus prior to any partitioning, with the combined pool split into training and validation subsets using an approximate 80/20 ratio applied to the full pooled set. Approximately 1,300 images were drawn randomly from each dataset prior to pooling, yielding a total combined pool of approximately 3,900 images, intentionally balanced across source datasets to prevent any single dataset from disproportionately shaping the learned decomposition structure. The same training and validation partitions are used for both the PCA and ICA pipelines to ensure that all comparative results reflect genuine methodological differences rather than differences in data exposure.

### B. OFIQ Metric Extraction

OFIQ is applied to all images in both the training and validation subsets via its command-line interface, OFIQSampleApp, producing a structured CSV output for each partition. The CSV structure comprises 58 columns: a filename column, 28 standard quality metrics, 28 scalar quality metrics, and a single unified quality score. The scalar metrics normalize measurement values onto a consistent range and are the only metrics fed into either decomposition pipeline. After excluding administrative fields and the unified quality score column — retained separately as the ground truth approximation target — 20 scalar metrics constitute the input feature space for PCA and 19 for ICA, the one-metric difference arising from the exclusion of an additional field identified as uninformative during ICA pre-processing.

A small number of images — between two and three across the full dataset — returned zero values across all metric columns due to OFIQ face detection failures and were excluded from further analysis without replacement.

### C. Variance-Threshold PCA Formulation

This study's primary methodological contribution is a variance-threshold-based PCA formulation that, without prescribing a fixed component count in advance, finds the minimum number of principal components needed to capture 95% of the cumulative explained variance in the OFIQ scalar metric space. The following steps are taken in the formulation process. To ensure that all features enter the decomposition on a common scale and remove the impact of different metric ranges and units on the covariance structure, all OFIQ scalar metric values in the training set are standardized using z-score normalization prior to PCA fitting:

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_i}$$

where $x_{ij}$ is the raw value of metric j for image i, $\mu_j$ is the mean of metric j across all training images, and $\sigma_j$ is the corresponding standard deviation. The standardization parameters $\mu_j$ and $\sigma_j$ are computed exclusively from the training set and saved for consistent application to the validation set, ensuring that no information from the validation partition influences the learned scaling boundaries. Following standardization, PCA is applied to the full matrix of scaled training metrics. The scaled data matrix Z is used to compute the covariance matrix C:

$$C = \frac{1}{1-n} Z^T Z$$

where n is the number of training images. The eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$ and their corresponding eigenvectors $v_1, v_2, \cdots, v_p$ are obtained by solving the eigenvalue equation:

$$Cv_k = \lambda_k v_k$$

where each eigenvalue $\lambda_k$ quantifies the proportion of total variance captured by its corresponding principal component and each eigenvector $v_k$ defines the direction of that component in the original feature space. The proportion of total variance explained by the k-th principal component is given by:

$$r_k = \frac{\lambda_k}{\sum_{j=1}^{p} \lambda_j}$$

The cumulative variance explained after retaining the first k components is then:

$$v_k = \sum_{i=1}^{k} r_i = \frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{k} \lambda_j}$$

The minimum number of components $k^*$ retained by the model is determined by identifying the smallest value of k for which the cumulative explained variance meets or exceeds the 95% threshold:

$$k^* = \min(k : v_k \geq 0.95)$$

This variance-threshold criterion allows the optimal component count to emerge organically from the distributional structure of the combined training data rather than being prescribed arbitrarily, ensuring that the retained components collectively capture a sufficient and consistent proportion of the total variance in the OFIQ metric space. The primary goal of the study is to demonstrate that $k^*$ is meaningfully smaller than the total number of OFIQ scalar metrics p, establishing that substantial dimensionality reduction is achievable while still satisfying the 95% cumulative variance criterion.

### D. Composite Score Construction

The projected score for each retained principal component is calculated for each image in the training and validation sets after component selection. The score on the k-th principal component for image i is determined by:

$$PC_{k,i} = Z_i v_k$$

where $Z_i$ is the standardized feature vector for image i and $v_k$ is the k-th eigenvector retained by the variance threshold criterion. The $k^*$ component scores are then combined into a single composite quality score using equal weighting across all retained components:

$$Q_i = \frac{1}{k^*} \sum_{k=1}^{k^*} PC_{k,i}$$

where $Q_i$ is the final composite quality score for image i. Equal weighting is applied across all retained components on the grounds that each component has been selected precisely because it contributes meaningfully to the 95% cumulative variance threshold, and no additional justification exists for differentially weighting components that have each met the same selection criterion. This composite score $Q_i$ serves as the PCA-based approximation of the OFIQ unified quality score and is used as the primary output of the model in the evaluation procedure.

### *E. FastICA Formulation*

Compared to PCA, Independent Component Analysis operates under a very distinct set of presumptions. ICA looks for components that are statistically independent from one another, which means that knowing the value of one component does not reveal anything about any other, as opposed to orthogonal components that maximize explained variance. The goal of ICA is to recover the underlying independent source signals, which are assumed to be linear blends of the observed scalar metrics. A variance threshold criterion cannot be directly applied to ICA since, in contrast to PCA, it does not generate an ordered set of components with a natural variance-based hierarchy. Rather, the FastICA algorithm's convergence behavior on the 19-metric input space after standardization determines the number of ICA components, which is set to 10.

This component count is lower than the 16 retained by PCA for two reasons: first, ICA operates on 19 rather than 20 metrics due to the exclusion of one additional uninformative field during ICA preprocessing; and second, FastICA's iterative convergence procedure identifies independent source directions that do not necessarily span the same dimensional space as PCA's variance-ordered components, often requiring fewer components to achieve stable convergence on datasets with moderate feature redundancy.

The ICA input metrics are subjected to the same z-score standardization as the PCA process, with parameters derived solely from the training partition. Then, using scikit-learn's FastICA with a set random seed for repeatability, FastICA is applied to the standardized training data to extract ten independent components. Component scores for each image are computed as:

$$IC_{k,i} = Z_i W_k^{ICA}$$

where $w_k^{ICA}$ is the k-th unmixing vector recovered by FastICA. Sign corrections are applied to components found to be negatively correlated with the OFIQ unified quality score. Equal weighting is applied across all 10 retained components, since ICA produces no natural variance-based ordering that would justify differential weighting:

$$Q_i^{ICA} = \frac{1}{10} \sum_{k=1}^{10} IC_{k,i}$$

Both composite scores are normalized to a 0–100 scale using min-max scaling boundaries fitted exclusively on the training partition and applied without refitting to the validation set.

### *F. Evaluation Procedure*

Both the PCA and ICA pipelines are applied to the held-out validation set using exclusively the parameters and model objects fitted during training — no refitting or recompilation is performed on the validation data. Performance is evaluated using three complementary metrics for each method: Pearson correlation coefficient between the composite score and the OFIQ unified quality score, measuring linear agreement; R-squared, measuring the proportion of variance in the OFIQ score explained by the composite; and mean absolute error (MAE) in score units on the 0–100 scale, measuring absolute deviation between the approximated and reference scores.

Individual per-metric Pearson correlations between each input scalar metric and the final composite score are also reported for both methods, providing insight into which quality dimensions most strongly drive each composite and whether the two methods weigh the underlying metric space differently.

## IV. Experimental Setup

The following section will walk through the full implementation of the methodology described in section III. We will first find images from the three datasets and divide them into an 80/20 training and validation set. Then we will install and implement the OFIQ software to run using the training dataset and generate overall quality values. Next, we will use the PCA algorithm to both determine the number of components needed, assess their correlations to the overall variance, and create the PCA components to calculate the new accuracy score. We will also do this with the ICA model to obtain its results. Lastly, we will use OFIQ to generate the overall quality values for the validation set and, using the PCA and ICA components, generate our own generated quality scores. These will then be compared to determine the accuracy of our algorithms, while also noting the decrease in computation resources required.

### *A. Dataset Acquisition and Pooling*

Images were downloaded from the official websites and repositories of VGGFace2, Flickr Faces HQ, and CelebA. All three datasets are distributed in large, compressed folder archives containing tens of thousands to hundreds of thousands of images, necessitating the downloading of substantial data volumes to access even a modest subset. Following download and extraction, approximately 1,300 images were selected at random from each dataset, yielding approximately equal per-dataset contributions to the combined pool. An example of the files we collected from the Flickr Faces HQ database is seen in figure 3.

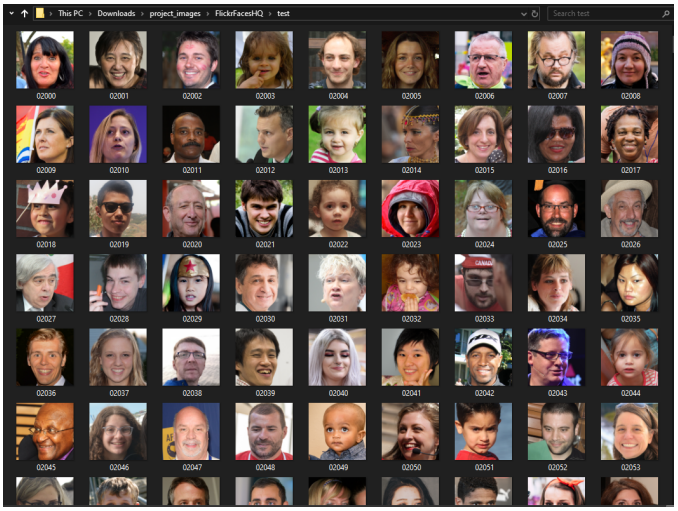

Fig. 3. FlickrFacesHQ Image Files for Training Dataset

This balance was intentional — ensuring that neither PCA nor ICA would be trained in a feature space dominated by the distributional properties of any single dataset. The three per-dataset subsets were merged into a unified pool of approximately 3,900 images, which was partitioned into a training set of 3,100 images and a validation set of 800 images using an approximate 80/20 split. The same partitions were used for both PCA and ICA training and validation. The combined pool was capped at approximately 3,900 images due to the computational overhead of running the full OFIQ pipeline at scale — preliminary testing indicated that processing more than 4,000 images resulted in prohibitively long execution times.

### B. OFIQ Initialization and Metric Extraction

By cloning the official OFIQ repository from GitHub, setting up the necessary dependencies, such as PyTorch and OpenCV, and executing the provided build scripts to produce the OFIQSampleApp command-line executable, OFIQ was installed on a local Windows computer. The OFIQSampleApp ran the complete OFIQ quality assessment pipeline on every image in each partition after being directed to the folders holding the downloaded training and validation photos.

The output from each partition was a structured CSV file, where each column represented a different output field and each row represented a single image. A filename column for identifying each image, 28 standard quality metrics for raw non-normalized quality measurements, 28 scalar quality metrics for corresponding normalized outputs calibrated for direct comparability across images and systems, and a single unified quality score column for OFIQ's overall composite assessment of each image's suitability for biometric recognition made up the CSV structure. Compared to the standard metrics, which collect raw measurement values that may differ significantly in size and range across images, the scalar metrics standardize measurement results into a uniform range, making

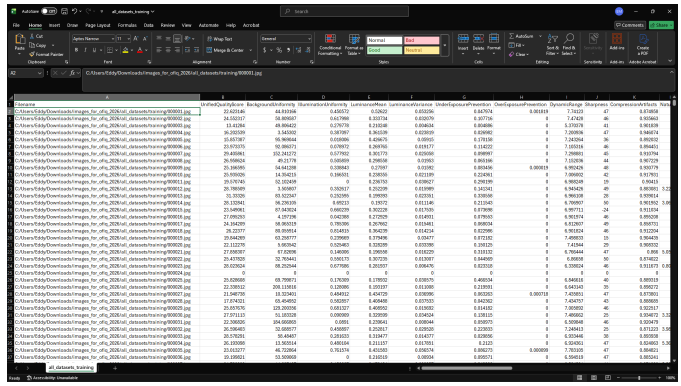

Fig. 4. OFIQ CSV Data after Processing

them more suitable for direct input into a dimensionality reduction pipeline. This can be seen in figure 4.

Following metric extraction, the scalar quality metrics were prepared as inputs for both the PCA and ICA pipelines independently. For the PCA pipeline, a pre-processing step excluded administrative fields such as assessment time and the unified quality score column — retained separately as the ground truth approximation target — reducing the input feature space from 27 scalar metrics to 20 and establishing $p_{PCA} = 20$ as the effective PCA input dimensionality against which the minimum component count $k^*$ is compared. For the ICA pipeline, the same exclusion procedure was applied with one additional metric removed following identification as uninformative during ICA-specific preprocessing, yielding a slightly different input feature space of 19 scalar metrics and establishing $p_{ICA} = 19$ as the ICA input dimensionality.

This one-metric difference between the two pipelines is a deliberate consequence of applying method-specific preprocessing criteria to each decomposition independently rather than enforcing identical inputs, and it is acknowledged as a minor asymmetry in the comparative analysis. The unified quality score was retained separately from both pipelines as the singular ground truth reference against which both composite scores are evaluated.

During OFIQ processing, a tiny number of photographs—between two and three in the entire combined pool—were found to have returned zero values across all metric columns, suggesting that the face detection phase for those specific images had failed. Since these images make up a very small percentage of the roughly 3,900 images in the combined pool, they were purposefully left out of the PCA and ICA pipelines without being replaced because their inclusion with zero-value metrics would cause artificial distortion in the covariance and independence structures that were used to fit both models.

### C. Data Standardization

To ensure that every feature entered the decomposition on a common scale independent of its initial measurement range, all scalar values in the training set were standardized using z-score normalization prior to fitting either the PCA or ICA

models. Both approaches require standardization; for PCA, it guarantees that features with greater absolute ranges do not disproportionately dominate the covariance structure and, consequently, the principal component directions; for ICA, it is equally important because FastICA's iterative convergence procedure assumes that input features have been pre-whitened and centered, and inconsistent scaling across features can either prevent the algorithm from converging reliably or cause it to recover spurious independent components. For the PCA pipeline, the mean $\mu_j$ and standard deviation $\sigma_j$ for each of the 20-input metrics were computed exclusively from the 3,100 training images. For the ICA pipeline, the same procedure was applied independently to the 19 ICA input metrics, again using only the training partition to compute standardization parameters. In both cases, the standardization parameters were saved alongside their respective trained model objects standard_scaler.joblib for PCA and ica_standard_scaler.joblib for ICA — ensuring that the validation set images are scaled using the identical reference boundaries established during training rather than their own distributional statistics.

This strict separation between training and validation scaling guarantees that no information from the validation partition influences the learned parameters of either model, or that the evaluation accurately reflects the true generalization capacity of each pipeline to unseen data. The saved standardization records further provide a reusable reference that can be applied to any future image collection processed through either pipeline without requiring recompilation, preserving the consistency and comparability of both PCA and ICA composite scores across deployment contexts.

### *D. PCA Training and Component Selection*

A Python script running in Google Colab was used to feed the standardized training metrics to both the PCA and ICA pipelines independently. For both methods, a pre-processing step was applied prior to fitting to exclude metrics deemed unrelated to the quality score approximation objective, including administrative fields such as assessment time and the OFIQ unified quality score column itself, which was retained separately as the ground truth target rather than used as a model input. For the PCA pipeline, this step reduced the input feature space from 27 scalar metrics to 20, establishing $p_{PCA} = 20$ as the effective dimensionality against which the minimum component count $k^*$ is compared. The PCA model was then fitted to the standardized training data using a variance threshold criterion of 95%, allowing the ideal component count to arise organically from the covariance structure of the data rather than being predetermined. The model retained $k^* = 16$ principal components, collectively capturing 96.55% of the cumulative explained variance and achieving a meaningful reduction from 20 to 16 dimensions.

The variance distribution across components is noticeably front-loaded, with PC1 and PC2 together accounting for approximately 34% of the total cumulative variance, after which contributions diminish progressively across the remaining components, consistent with the existence of a dominant

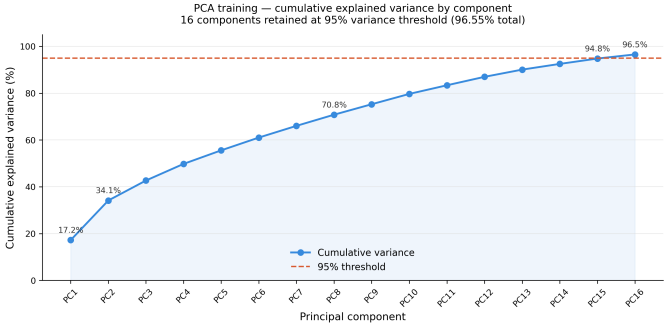


Fig. 5. PCA Cumulative Variance per Component

axis of quality-related variation in the OFIQ scalar metric space. This can be seen in figure 5.

Variance-proportional weights were assigned to each retained component, with each component's weight calculated as its individual explained variance divided by the total explained variance across all 16 retained components, ensuring that higher-variance components exert proportionally greater influence over the final composite quality score.

A sign correction step was additionally applied to each component by examining its correlation with the OFIQ unified quality score reference — components found to be negatively correlated with the reference were sign-flipped prior to weighted combination to ensure directional consistency. Specifically, PC1, PC2, PC3, PC8, PC9, PC11, PC12, PC14, and PC16 were sign-flipped, while PC4, PC5, PC6, PC7, PC10, PC13, and PC15 required no correction. The trained PCA object, standardization parameters, component weights, and sign correction factors were saved to the output directory for subsequent application to the validation set across six model files: full_pca_model.joblib, standard_scaler.joblib, minmax_scaler_full.joblib, component_weights.joblib, metric_weights.joblib, and sign_corrections.joblib.

For the ICA pipeline, the same exclusion procedure was applied with one additional metric removed during ICA-specific pre-processing, reducing the input feature space to 19 scalar metrics and establishing $P_{ICA} = 19$ as the effective ICA input dimensionality. FastICA was run on the standardized 19-metric training data with a maximum of 1,000 iterations and successfully converged, extracting 10 independent components. Notably, all 10 ICA component correlations with the OFIQ unified quality score are substantially lower in magnitude than the leading PCA component correlations, reflecting the fundamental difference between the two methods — ICA components are not ordered by variance explained and do not necessarily align with the dominant axes of quality variation in the metric space.

Equal weights of 0.10 were assigned to each of the 10 components given ICA's absence of a natural variance-based ordering that would justify differential weighting. The combined ICA composite score was normalized to a 0–100 scale using a MinMaxScaler fitted exclusively on the training data. Five ICA model files were saved to the output

directory for validation set application: ica_model.joblib, ica_standard_scaler.joblib, ica_minmax_scaler.joblib, ica_sign_correction.joblib, and ica_n_components.joblib.

### *E. Composite Score Generation*

For the PCA pipeline, the saved PCA object, standardization parameters, component weights, and sign correction factors were applied to the validation set without any refitting or recompilation on the validation data itself, ensuring that the evaluation accurately reflects the true generalization capacity of the model learned solely from the training partition. The validation metrics were scaled to the same feature space as the training data using the training-fitted standardization parameters. The scaled metric vector of each validation image was then projected into the PCA component space using the retained 16 eigenvectors, producing a variance-proportionally weighted composite quality score for each image. Two complementary dimensions were used to assess PCA performance.

First, the overall Pearson correlation coefficient between the PCA composite scores and the OFIQ unified quality score was computed across all validation images to measure overall approximation fidelity. Second, individual Pearson correlations between each of the 20 input scalar metrics and the PCA composite score were computed, providing insight into which quality dimensions most strongly drive the composite and whether the model's weighting behavior is consistent with the established determinants of facial image quality. The most strongly correlated individual metric during training was FaceOcclusionPrevention.scalar at +0.7233, followed by the OFIQ unified quality score at +0.6728, suggesting meaningful alignment between the composite score and the reference target even at the individual metric level.

For the ICA pipeline, the same procedure was applied in parallel — the five saved ICA model files were loaded, and the ICA transform was applied to the standardized 19-metric validation data without refitting. Sign corrections were applied and the 10 equal-weighted independent component scores were combined and normalized to the 0–100 scale using the training-fitted MinMaxScaler. Individual Pearson correlations between each of the 19 input scalar metrics and the ICA composite score were computed alongside the overall correlation with the OFIQ unified quality score, enabling a direct per-metric comparison with the PCA results. Full quantitative results and a detailed comparative discussion of both methods' findings are presented in Section V.

### *F. Validation Set Processing*

The validation phase for both pipelines was initiated by uploading the validation OFIQ metrics CSV alongside all saved model files from both the PCA and ICA training phases into the Google Colab environment. For the PCA pipeline, the six saved model files collectively encode the complete learned state of the PCA pipeline as follows: full_pca_model.joblib stores the fitted PCA object containing the 16 retained eigenvectors and their associated eigenvalues; standard_scaler.joblib stores the z-score standardization parameters computed from the training set; minmax_scaler_full.joblib stores the min-max scaling boundaries used to normalize composite scores to the 0–100 range; component_weights.joblib stores the variance-proportional weights assigned to each of the 16 retained components; metric_weights.joblib stores input metric weighting factors applied prior to standardization; and sign_corrections.joblib stores the directional correction factors applied to each component.

The pipeline applied metric weighting, standardization, PCA projection, sign correction, and variance-proportional weighted combination sequentially to the 1,168 validation images, producing a composite quality score on a 0–100 scale for each image. The PCA validation score distribution produced a mean of 86.07, a standard deviation of 8.06, and a range of 0 to 99.27, closely mirroring the training distribution (mean 85.64, standard deviation 8.49). The OFIQ unified quality score maintained a correlation of +0.6421 with the composite PCA output on the validation set, a modest decline from the training correlation of +0.6728, with FaceOcclusionPrevention.scalar remaining the most strongly correlated individual metric at +0.7117.

For the ICA pipeline, the five saved model files were loaded and applied to the same 1,168 validation images using the 19-metric input space. The ICA pipeline applied standardization, ICA transform, sign correction, equal-weighted combination, and min-max scaling sequentially, using the existing MinMaxScaler without refitting. The ICA validation score distribution produced a mean of 91.10, a standard deviation of 5.75, and a range of 0 to 100.00. The OFIQ unified quality score maintained a correlation of +0.6421 with the PCA composite and +0.3753 with the ICA composite across the validation set. Section V presents the complete quantitative analysis and comparative interpretation of both pipelines' validation results.

### *G. Performance Evaluation*

Following the application of both pipelines to the validation set, the composite quality scores generated for all 1,168 validation images by each method were evaluated against the OFIQ unified quality score using three performance metrics: Pearson correlation coefficient, R-squared, and mean absolute error in score units on the 0–100 scale.

For the PCA pipeline, the Pearson correlation between the composite score and the OFIQ unified quality score was +0.6421, the R-squared was 0.4124, and the MAE was 36.57 units. The PCA composite score distribution exhibited a mean of 86.07 and standard deviation of 8.06, closely consistent with both the PCA training distribution and the OFIQ reference distribution, indicating stable generalization without meaningful distributional shift. Metrics reflecting occlusion, illumination, and facial framing — including FaceOcclusionPrevention.scalar, MouthOcclusionPrevention.scalar, EyesVisible.scalar, and UnderExposurePrevention.scalar — maintained broadly consistent and strong correlation rankings between training and validation, while BackgroundUniformity.scalar and Sharpness.scalar exhibited the weakest correlations with the PCA composite score in both phases.

For the ICA pipeline, the Pearson correlation between the composite score and the OFIQ unified quality score was +0.3753, the R-squared was 0.1408, and the MAE was 41.56 score units. The ICA composite score distribution exhibited a mean of 91.10 and standard deviation of 5.75. Metrics exhibiting the strongest correlations with the ICA composite score on the validation set included CompressionArtifacts.scalar at +0.6592, UnderExposurePrevention.scalar at +0.5531, FaceOcclusionPrevention.scalar at +0.5713, and OverExposurePrevention.scalar at +0.4616, while LuminanceMean.scalar at +0.1716 and MarginBelowOfTheFaceImage.scalar at -0.1705 exhibited the weakest correlations. A comprehensive comparative analysis and interpretation of the PCA and ICA results, including discussion of the performance gap between the two methods and the per-metric correlation patterns that illuminate its causes, is presented in Section V.

## V. Results and Analysis

The results of this study demonstrate a clear and consistent performance advantage for PCA over ICA across all three evaluation metrics on the validation set. The Pearson correlation between the PCA composite score and the OFIQ unified quality score reached +0.6421 on the validation set, compared to +0.3753 for the ICA composite — a difference of 0.2668 correlation points representing a substantial and practically meaningful gap in approximation fidelity. The R-squared values reinforce this finding, with PCA explaining 41.24% of the variance in the OFIQ unified quality score against ICA's 14.08%, indicating that the PCA composite captures nearly three times as much of the reference score's variance as its ICA counterpart. The MAE results are similarly consistent, with PCA producing a lower absolute error of 36.57 score units compared to ICA's 41.56 — a difference of approximately 5 points on the 0–100 scale that compounds meaningfully at operational dataset sizes. Across all three metrics, PCA outperforms ICA, and the comparative charts in figures 6 and 7 summarize these findings visually.

The per-metric correlation analysis provides deeper insight into why PCA outperforms ICA overall while also revealing specific quality dimensions where ICA demonstrates superior sensitivity. PCA exhibits stronger correlations with occlusion-related, geometric, and facial framing metrics — FaceOcclusionPrevention.scalar at +0.7117, EyesVisible.scalar at +0.4789, HeadSize.scalar at +0.4334, and the margin metrics — suggesting that the dominant axes of variance in the OFIQ scalar metric space are largely driven by these quality dimensions, which PCA is well-positioned to capture. ICA, by contrast, exhibits notably stronger correlations with metrics reflecting image-level photographic properties — CompressionArtifacts.scalar at +0.6592 versus PCA's +0.4467, OverExposurePrevention.scalar at +0.4616 versus PCA's +0.2629, Sharpness.scalar at +0.3670 versus PCA's +0.0098, and NaturalColour.scalar at +0.3585 versus PCA's +0.1518.

This pattern is consistent with ICA's underlying assumption that the observed metrics are mixtures of statistically independent latent factors — the photographic quality dimensions captured more strongly by ICA may reflect genuinely independent underlying sources of image variation that PCA's variance-maximization objective does not prioritize.

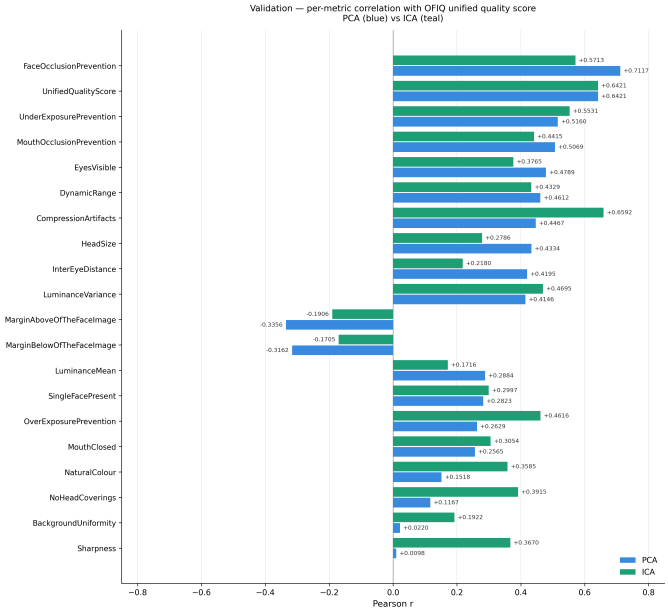


Fig. 6. PCA Vs ICA per Metric Correlation

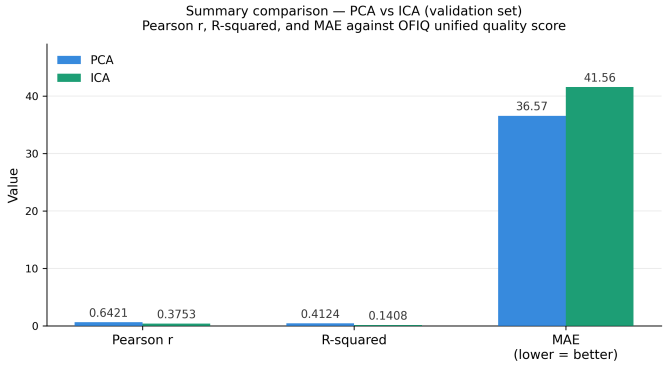


Fig. 7. PCA Vs ICA Pearson, R-Squared, and MAE Comparison

The score distribution comparison also reveals meaningful structural differences between the two methods. The PCA validation distribution (mean 86.07, standard deviation 8.06) closely mirrors the training distribution (mean 85.64, standard deviation 8.49), indicating stable generalization. The ICA validation distribution (mean 91.10, standard deviation 5.75) is notably compressed and skewed toward the upper end of the 0–100 scale relative to the OFIQ reference distribution, suggesting that ICA's equal-weighted combination of independent components produces a composite that does not span the full quality score range as effectively as the PCA composite. This distributional mismatch likely contributes to ICA's higher MAE, as its scores systematically cluster toward higher values even for images the OFIQ unified quality score rates more modestly.

Regarding dimensionality reduction, the PCA model retained k*=16 components from p=20 metrics, achieving a 20% reduction while capturing 96.55% of cumulative explained variance. The ICA model retained 10 components from 19 metrics — a 47% reduction in component count relative to the input space — though this reduction is achieved at a substantially greater cost to approximation fidelity as demonstrated by the comparison metrics above.

For large-scale biometric deployments where computational efficiency is the primary concern and approximation fidelity requirements are moderate, ICA's more aggressive dimensionality reduction may offer practical value. However, where fidelity to the OFIQ unified quality score is the primary objective, PCA's 16-component composite demonstrates meaningfully superior performance across all evaluated metrics and should be preferred. Both methods' saved model files enable lightweight reusable approximation without re-executing the full OFIQ pipeline on new image collections, representing a shared practical advantage over the full pipeline for large-scale quality-gated acquisition contexts.

## VI. Conclusion

This study demonstrated that both PCA and ICA can approximate the OFIQ unified quality score from a compact set of scalar metric inputs, with PCA consistently outperforming ICA across all three-evaluation metrics on the combined validation set. The PCA composite achieved a Pearson correlation of +0.6421, R-squared of 0.4124, and MAE of 36.57, compared to ICA's +0.3753, 0.1408, and 41.56 respectively - a performance gap attributable to PCA's alignment with the dominant axes of variance in the OFIQ metric space, which are primarily driven by occlusion, geometric, and facial framing quality dimensions. ICA demonstrated superior sensitivity to photographic quality dimensions including compression artifacts, sharpness, and exposure, suggesting that the two methods capture complementary aspects of the underlying quality metric space and that a hybrid approach combining both decompositions may warrant investigation in future work.

The variance-threshold PCA criterion successfully identified $k^* = 16$ components as sufficient to capture 96.55% of cumulative explained variance from 20 input metrics, achieving a 20% dimensionality reduction while preserving most of the metric space's informational content. Both pipelines produce reusable saved model files enabling lightweight quality approximation on new image collections without re-executing the full OFIQ pipeline, offering meaningful computational efficiency gains for large-scale biometric deployments. Future work should explore hybrid PCA-ICA formulations, higher variance thresholds, and extended validation across additional demographically diverse datasets to further improve approximation fidelity and dimensionality reduction simultaneously.

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